



GOSFORD HIGH SCHOOL

2008

YEAR 12 HALF YEARLY HIGHER SCHOOL CERTIFICATE

MATHEMATICS

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question.

Total marks: - 100

- Attempt Questions 1 - 8
- Questions are worth 12 or 13 marks each

2008 Year 12 Half Yearly Examination

Question 1 (13 Marks)

Marks

a) Simplify $\frac{6a^2 - 9a}{3 - 2a}$ 2

b) Find the value of $\frac{mn}{m - n}$ when $m = 6.92 \times 10^5$ and $n = 7.9 \times 10^4$ 2
correct to 3 significant figures.

c) Solve $|3y - 2| < 7$ 2

d) Solve $(x + 3)(x - 1) = 32$ 2

e) If $2 \cos^2 \theta = 1$, find θ for $-180^\circ \leq \theta \leq 180^\circ$ 2

f) An arc AB of a sector of a circle is of length $\frac{\pi}{4}$ metres, and subtends an angle of 30° at the centre O of the circle.

(i) Show that the exact length of the radius of the circle is 1.5 metres 1

(ii) Find the area of the sector AOB 2

Question 2 (12 Marks) Begin a New Booklet

a) Solve $2x^2 - x - 3 \geq 0$ 2

b) For the quadratic equation $2x^2 - 9x + k = 0$, find:

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$

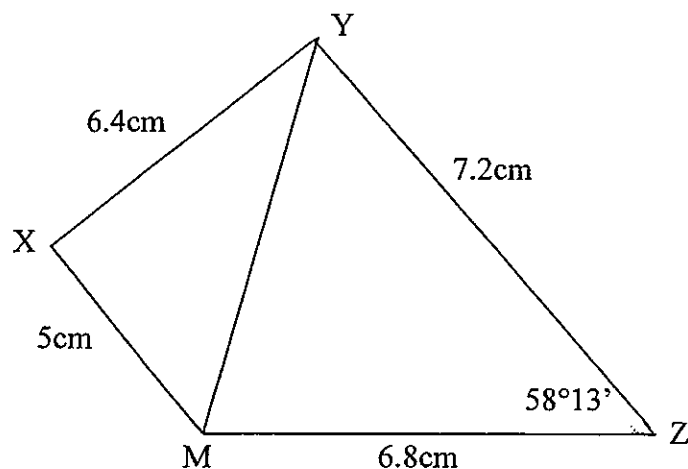
(iii) The value of k if the roots differ by 1 1

Question 2 continues on the next page

Question 2 continued

Marks

c)

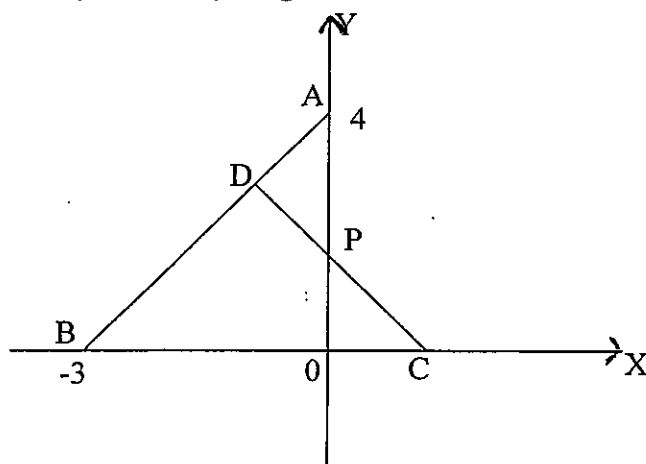


NOT TO SCALE

XYZM is a quadrilateral in which
 $XY=6.4\text{cm}$, $YZ=7.2\text{cm}$, $ZM=6.8\text{cm}$
 and $XM=5\text{cm}$. $\angle YZM=58^\circ13'$

- Show that YM is 6.8 cm correct to 1 decimal place. 2
- Show that $\angle MXY = 72^\circ3'$ correct to the nearest minute. 2
- Find the area of the quadrilateral XYZM correct to the nearest cm^2 2

Question 3 (12 Marks) Begin a new booklet.



NOT TO SCALE

In the diagram, $AB=BC$
 and CD is perpendicular to AB

CD intersects the Y axis at P

- Find the length of AB 1
- Show that the coordinates of C are $(2, 0)$ 1
- Show that the equation of CD is $3x + 4y = 6$ 2
- Find the coordinates of P 1
- Find the length of CP 2
- Prove that $\triangle ADP$ is congruent to $\triangle COP$ 3
- Hence, calculate the area of quadrilateral $DPOB$ 2

Question4 (13 Marks) Begin a new booklet

a) Differentiate with respect to x :

(i) $\frac{1}{x\sqrt{x}}$ 2

(ii) $(4 - 3x)^6$ 2

(iii) $\frac{\log x^2}{x}$ 2

b) Evaluate $\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$ 2

c) Write down the quadratic equation whose roots are $1 + \sqrt{3}$ and $1 - \sqrt{3}$ 2

d) (i) Find the coordinates of the centre of the circle

$$x^2 + y^2 + 4x - 2y = 11$$
 2

(ii) State the domain of this circle 1

Question 5 (13 Marks) Begin a new booklet

a) Find $\int \frac{4x}{x^2 + 1} dx$ 1

b) Evaluate $\int_0^1 e^{-3x} dx$ correct to 3 decimal places 2

c) The equation of a parabola is given by $12y = x^2 - 6x - 15$

(i) Write this equation in standard form 1

(ii) Sketch the parabola 1

(iii) Find the coordinates of the vertex 1

(iv) Find the coordinates of the focus 1

d) For a particular curve, $\frac{d^2y}{dx^2} = 2$ for all points on the curve.

Find y in terms of x if $\frac{dy}{dx} = 10$ and $y = 3$ when $x = 0$ 3

e) A and B are the points (1, 0) and (9, 0) respectively. P (x, y) is the point such that

$PA^2 + PO^2 = PB^2$, where O is the origin. Show that the locus of P is

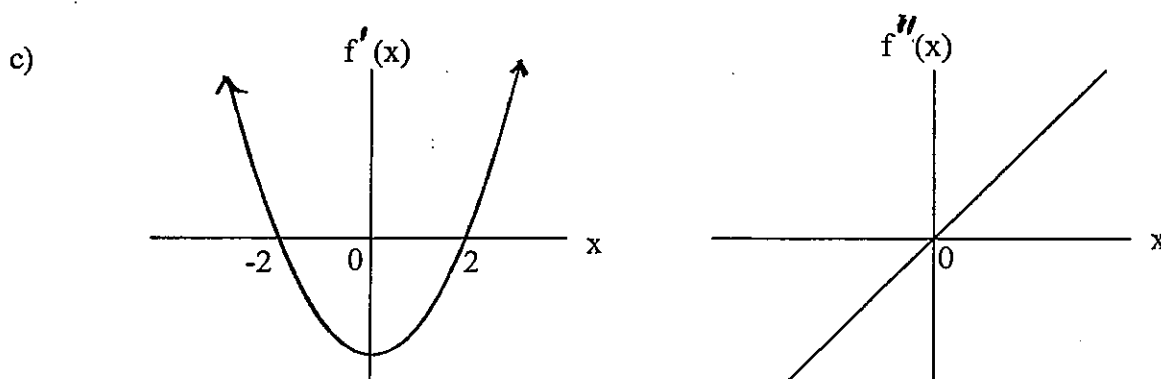
$$x^2 + y^2 + 16x - 80 = 0$$
 3

Question 6 (12 Marks) Start a new booklet

- a) For the curve $y = x^3 - 3x^2 - 9x + 10$
- (i) Find the stationary points 2
 - (ii) Determine their nature 2
 - (iii) Find the point of inflexion 2
 - (iv) Sketch the curve in the domain $-2 \leq x \leq 4$ 3
- b) Find the equation of the normal to the curve $y = x^3 - 3x^2 - 9x + 10$ at the point where $x = 0$. 3

Question 7 (13 Marks) Begin a new booklet

- a) (i) Differentiate $x.e^{2x}$ 2
- (ii) Hence, evaluate $\int_0^2 2x.e^{2x} dx$ as an exact value 3
- b) The area under the curve $y = e^x - 1$ from $x = 0$ to $x = 2$, is rotated about the X-axis. Find the exact volume generated. 3



These graphs show the first and second derivatives of a curve $y = f(x)$.
For which values of x is the function:

- (i) increasing 1
 - (ii) concave down? 1
- d) Use Simpson's Rule with 5 function values to estimate the value of

$$\int_1^5 \frac{x^2}{x^2 + 1} dx \quad \text{correct to 2 decimal places} \quad \text{3}$$

Question 8 (12 Marks) Begin a new booklet

a) If $\log_e 2 = x$ and $\log_e 3 = y$, express in terms of x and y :

(i) $\log_e 0.25$ **1**

(ii) $\log_e \left(\frac{8}{9} \right)$ **1**

b) Solve for x : $2x \log_a 4 = \log_a 8$ (a is a positive constant) **2**

c) Find the area between the curve $y = \log_e x$, the Y axis and the lines $y = 1$, and $y = 3$. (Answer to 3 significant figures) **2**

d) A cylinder of radius r and height h is **open at one end**.

(i) If the **volume** of the cylinder is to be 1000 cm^3 , find an expression for h in terms of r . **1**

(ii) Show that the **surface area** of this open cylinder is given by

$$A = \pi r^2 + \frac{2000}{r} \quad \text{2}$$

(iii) Find the radius of the cylinder with least surface area. **3**

Give an exact answer, then an approximation to 2 significant figures

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

Year 12 1/2 July 2008
Question 1 (13 marks)

a) $3a(2a-3) = -3a$

b) $\frac{(6.92 \times 10^5)(7.9 \times 10^4)}{6.92 \times 10^5 - 7.9 \times 10^4} = 89161.077$
 $= 89,200$ (3 s.f.)

c) $3y-2 < 7$ OR $3y-2 > -7$
 $3y < 9$ $3y > -5$
 $y < 3$ $y > -\frac{5}{3}$
 $-\frac{5}{3} < y < 3$

d) $x^2+2x-3 = 32$
 $x^2+2x-35 = 0$
 $(x+7)(x-5) = 0$
 $\therefore x = -7$ OR $x = 5$

e) $2\cos^2\theta = 1$
 $\cos^2\theta = \frac{1}{2}$
 $\cos\theta = \pm \frac{1}{\sqrt{2}}$
 $\therefore \theta = -135^\circ, -45^\circ, 45^\circ, 135^\circ$

f) i) $\ell = r\theta$ ($30^\circ = \frac{\pi}{6}$)
 $\frac{\pi}{4} = r \times \frac{\pi}{6}$
 $r = 1.5$

$\therefore$ radius is 1.5 m.

ii) $A = \frac{1}{2}r^2\theta$
 $A = \frac{1}{2} \times (1.5)^2 \times \frac{\pi}{6}$
 $A = \frac{3\pi}{16}$
 $\therefore$ Area is $\frac{3\pi}{16} \text{ m}^2$

Question 2 (12 marks)

a) For $(2x-3)(x+1) = 0$
 $x = \frac{3}{2}$ OR $x = -1$
Test $x = 0$ in $2x^2 - x - 3 \geq 0$
 $-3 \geq 0$ F.
 $\therefore x \leq -1$ and $x \geq \frac{3}{2}$

Q2 CONT'D

b) $2x^2 - 9x + k = 0$

i) $\alpha + \beta = -\frac{b}{a}$
 $= \frac{9}{2}$

ii) $\alpha\beta = \frac{c}{a}$
 $= \frac{k}{2}$

iii) Roots are α and $(\alpha+1)$

$\alpha + (\alpha+1) = \frac{9}{2}$

$2\alpha = \frac{7}{2}$

$\alpha = \frac{7}{4}$

Roots are $\frac{7}{4}$ and $\frac{11}{4}$

Now $\frac{7}{4} \times \frac{11}{4} = \frac{k}{2}$
 $k = \frac{77}{8}$

c) i) $YM^2 = 6.8^2 + 7.2^2 - 2 \times 6.8 \times 7.2 \times \cos 58^\circ 13'$

$YM = 6.8 \text{ cm}$ (1 d.p.) as required

ii) $\cos \angle MYX = \frac{5^2 + 6.4^2 - 6.8^2}{2 \times 5 \times 6.4}$

$\angle MYX = 72^\circ 3'$ (as required)

iii) Area $\triangle XYM$: $A = \frac{1}{2} \times 5 \times 6.4 \times \sin 72^\circ 3'$

Area $\triangle YZM$: $A = \frac{1}{2} \times 7.2 \times 6.8 \times \sin 58^\circ 13'$

$\therefore$ Area quad $XYZM = 36 \text{ cm}^2$ (nearest cm^2)

Question 3 (12 marks)

a) $AB^2 = 3^2 + 4^2$

$AB = 5$ units

b) $AB = BC$ (given)

$AB = 5$ units (shown in (a))

$\therefore BC = 5$ units

Hence $C = (2, 0)$

c). Grad $AB = \frac{4}{3}$

Grad $CD = -\frac{3}{4}$

Eqn of CD : $y - y_1 = m(x - x_1)$

$y - 0 = -\frac{3}{4}(x - 2)$

$4y = -3x + 6$

$\therefore 3x + 4y = 6$ as req'd.

d). When $x = 0$, $0 + 4y = 6$

$y = \frac{3}{2}$

Q3 (CONT'D)

$$c) CP^2 = 2^2 + \left(\frac{3}{2}\right)^2$$

$$= 4 + \frac{9}{4}$$

$$CP = \sqrt{\frac{25}{4}}$$

$$\therefore CP = 2\frac{1}{2} \text{ units}$$

f) Aim: To prove $\triangle ADP \equiv \triangle COP$

Proof: In $\triangle ADP, COP$

$$\angle ADP = \angle COP (90^\circ)$$

$$\angle DPA = \angle OPC (\text{vertically opp.})$$

$$\text{Now } PA = (4 - 1\frac{1}{2}) \text{ units}$$

$$= 2\frac{1}{2} \text{ units}$$

$$\text{and } CP = 2\frac{1}{2} \text{ units (shown in e)}$$

$$\therefore PA = PC$$

$$\text{Hence } \triangle ADP \equiv \triangle COP (AAS)$$

$$g) \text{ Area } \triangle ABO = 6 \text{ units}^2$$

$$\text{Area } \triangle COP = \frac{1}{2} \times \frac{3}{2} \times 2 \text{ units}^2$$

$$= 1\frac{1}{2} \text{ units}^2$$

$$\text{Now } \triangle COP \equiv \triangle ADP$$

$$\therefore \text{Area } \triangle ADF = 1\frac{1}{2} \text{ units}^2$$

$$\text{Hence quad. } DPOB = 4\frac{1}{2} \text{ units}^2$$

Question 4 (13 marks)

$$a) i) \frac{d}{dx} (x^{-\frac{3}{2}}) = -\frac{3}{2} x^{-\frac{5}{2}}$$

$$= -\frac{3}{2\sqrt{x^5}} \text{ OR } -\frac{3}{2x^2\sqrt{x}}$$

$$ii) \frac{d}{dx} (4-3x)^6 = 6(4-3x)^5 \times -3$$

$$= -18(4-3x)^5$$

$$iii) \frac{d}{dx} \left(\frac{\log x^2}{x} \right) = \frac{d}{dx} \left[\frac{2 \log x}{x} \right]$$

$$= \frac{x \cdot \frac{2}{x} - 2 \log x \cdot 1}{x^2}$$

$$= \frac{2 - 2 \log x}{x^2}$$

$$b) \lim_{x \rightarrow -1} \frac{x^3+1}{x+1} = \lim_{x \rightarrow -1} \frac{(x+1)(x^2-x+1)}{(x+1)}$$

$$= \frac{(-1)^2 + 1 + 1}{1}$$

$$= 3$$

Q4 CONT'D

$$c) x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - (1+\sqrt{3} + 1-\sqrt{3})x + (1+\sqrt{3})(1-\sqrt{3}) = 0$$

$$x^2 - 2x + 1 - 3 = 0$$

$$\text{i.e. } x^2 - 2x - 2 = 0$$

$$d) i) x^2 + y^2 + 4x - 2y = 11$$

$$(x^2 + 4x) + (y^2 - 2y) = 11$$

$$(x^2 + 4x + 4) + (y^2 - 2y + 1) = 11 + 4 + 1$$

$$(x+2)^2 + (y-1)^2 = 16$$

$$\therefore \text{Centre} = (-2, 1)$$

$$ii) \text{Radius} = 4$$

$$\therefore \text{Domain is } \{x: -6 \leq x \leq 2\}$$

Question 5 (13 marks)

$$a) \int \frac{4x}{x^2+1} dx = 2 \int \frac{2x}{x^2+1} dx$$

$$= 2 \log(x^2+1) + C$$

$$b) \int_0^1 e^{-3x} dx = \left[-\frac{1}{3} e^{-3x} \right]_0^1$$

$$= -\frac{1}{3} (e^{-3} - 1)$$

$$= 0.317 \text{ (3 dp)}$$

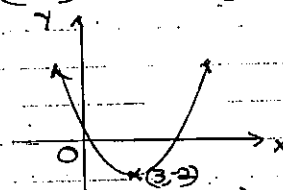
$$c) 12y = x^2 - 6x - 15$$

$$i) x^2 - 6x = 12y + 15$$

$$x^2 - 6x + 9 = 12y + 24$$

$$(x-3)^2 = 12(y+2)$$

ii)



$$iii) \text{Vertex} = (3, -2)$$

$$iv) \text{Focus} = (3, 1)$$

$$d) \frac{d^2y}{dx^2} = 2$$

$$\frac{dy}{dx} = 2x + C$$

$$\text{when } x=0, \frac{dy}{dx} = 10$$

$$\therefore C = 10$$

$$\text{Hence } \frac{dy}{dx} = 2x + 10$$

$$\text{Now } y = x^2 + 10x + C$$

$$\text{when } x=0, y=3$$

$$3 = C$$

$$\therefore y = x^2 + 10x + 3$$

Q5 CONT'D

e) Dist PA = $\sqrt{(x-1)^2 + (y-0)^2}$

" PO = $\sqrt{(x-0)^2 + (y-0)^2}$

" PB = $\sqrt{(x-9)^2 + (y-0)^2}$

Now $PA^2 + PO^2 = PB^2$

$\therefore (x-1)^2 + y^2 + x^2 + y^2 = (x-9)^2 + y^2$

$x^2 - 2x + 1 + y^2 + x^2 + y^2 = x^2 - 18x + 81 + y^2$

$x^2 + 16x + y^2 - 80 = 0$

$\therefore$ The locus of P is $x^2 + y^2 + 16x - 80 = 0$

3

Question 6 (12 marks)

a) $y = x^3 - 3x^2 - 9x + 10$

i) $\frac{dy}{dx} = 3x^2 - 6x - 9$

$\frac{d^2y}{dx^2} = 6x - 6$

sp. occur when $\frac{dy}{dx} = 0$

$3(x^2 - 2x - 3) = 0$

$3(x-3)(x+1) = 0$

$\therefore x = 3 \text{ OR } x = -1$

when $x = 3, y = -17$

when $x = -1, y = 15$

$\therefore$ Stationary points are $(3, -17)$ & $(-1, 15)$

2

ii) when $x = 3, \frac{d^2y}{dx^2} > 0$

when $x = -1, \frac{d^2y}{dx^2} < 0$

$\therefore (3, -17)$ is a minimum t.p

and $(-1, 15)$ is a maximum t.p

2

iii) Possible point of inflexion occurs when $\frac{d^2y}{dx^2} = 0$

$6x - 6 = 0$

$x = 1$

when $x = 1, y = -1$

For $x < 1, \frac{d^2y}{dx^2} < 0$

For $x > 1, \frac{d^2y}{dx^2} > 0$

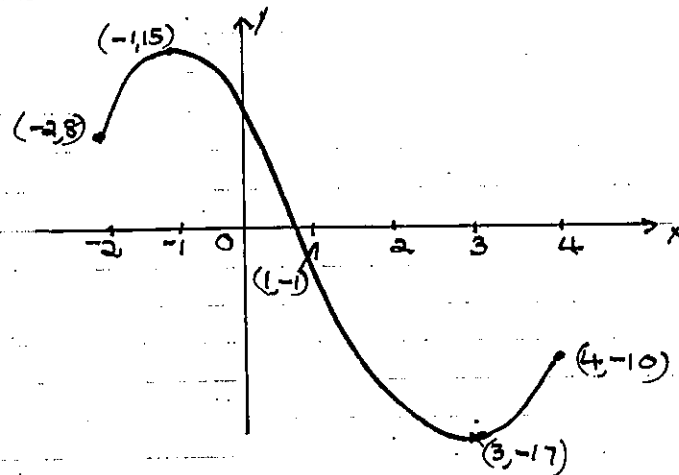
A change in concavity occurs

$\therefore (1, -1)$ is an inflexion point

2

iv) When $x = -2, y = 8$

Q6 CONT'D



3

b) $y = x^3 - 3x^2 - 9x + 10$

$\frac{dy}{dx} = 3x^2 - 6x - 9$

when $x = 0, \frac{dy}{dx} = -9$

$m_{\text{tangent}} = -9$

$m_{\text{normal}} = \frac{1}{9}$

when $x = 0, y = 10$

Eqn of normal:

$y - y_1 = m(x - x_1)$

$y - 10 = \frac{1}{9}(x - 0)$

$9y - 90 = x$

$x - 9y + 90 = 0$

3

Question 7 (13 marks)

a) i) $\frac{d}{dx}(x.e^{2x}) = 1.e^{2x} + x.2e^{2x}$

$= e^{2x}(1 + 2x)$

OR $e^{2x} + 2x.e^{2x}$

2

ii) $\int_0^2 2x.e^{2x} dx = \left[x.e^{2x} - \frac{1}{2}e^{2x} \right]_0^2$

$= (2e^4 - \frac{1}{2}e^4) - (0 - \frac{1}{2}e^0)$

$= 2e^4 - \frac{1}{2}e^4 + \frac{1}{2}$

$= \frac{3e^4 + 1}{2}$

3

Q7. CONT'D

$$\begin{aligned} y &= e^x - 1 \\ y^2 &= (e^x - 1)^2 \\ y^2 &= e^{2x} - 2e^x + 1 \\ \therefore V &= \pi \int_0^2 (e^{2x} - 2e^x + 1) dx \end{aligned}$$

$$V = \pi \left[\frac{1}{2} e^{2x} - 2e^x + x \right]_0^2$$

$$V = \pi \left[\left(\frac{1}{2} e^4 - 2e^2 + 2 \right) - \left(\frac{1}{2} - 2 + 0 \right) \right]$$

$$V = \pi \left[\frac{1}{2} e^4 - 2e^2 + 2 - \frac{1}{2} + 2 \right]$$

$$V = \pi \left(\frac{1}{2} e^4 - 2e^2 + 3\frac{1}{2} \right)$$

$$\text{OR } V = \frac{\pi}{2} (e^4 - 4e^2 + 7) \text{ units}^3 \quad 3$$

c) i) Increasing for $x < -2$, $x > 2$

ii) Concave down when $x < 0$

d)

x	1	2	3	4	5
f(x)	$\frac{1}{2}$	$\frac{4}{5}$	$\frac{9}{10}$	$\frac{16}{25}$	$\frac{25}{32}$

$$A \approx \frac{1}{3} \left\{ \frac{1}{2} + \frac{25}{26} + 2 \times \frac{9}{10} + 4 \left(\frac{4}{5} + \frac{16}{17} \right) \right\}$$

$$A = 3.41 \text{ (2 d.p.)} \quad 3$$

Question 8. (2 marks)

a) $\log_e 2 = x$ $\log_e 3 = y$

i) $\log_e 0.25 = \log_e 2^{-2}$
 $= -2 \log_e 2$
 $= -2x$

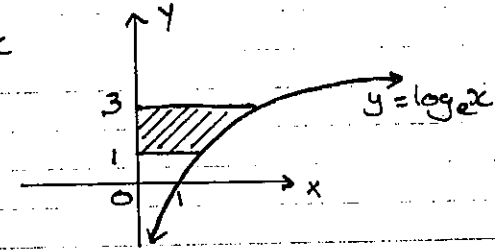
ii) $\log_e \frac{8}{9} = \log_e 8 - \log_e 9$
 $= 3 \log_e 2 - 2 \log_e 3$
 $= 3x - 2y$

b) $2x \log_e 4 = \log_e 8$
 $2x \cdot 2 \log_e 2 = 3 \log_e 2$
 $4x \log_e 2 = 3 \log_e 2$
 $4x = 3$
 $x = \frac{3}{4}$

Q8. CONT'D

c) $y = \log_e x$

$$x = e^y$$



$$A = \int_1^3 e^y dy$$

$$A = [e^y]_1^3$$

$$A = e^3 - e$$

$$A = 17.4 \text{ (3 s.f.)} \quad 2$$

d) i) $V = \pi r^2 h$

$$1000 = \pi r^2 h$$

$$\therefore h = \frac{1000}{\pi r^2}$$

ii) $SA = \pi r^2 + 2\pi r h$

$$SA = \pi r^2 + 2\pi r \times \frac{1000}{\pi r^2}$$

ie. $A = \pi r^2 + \frac{2000}{r}$ as req'd. 2

iii) $A = \pi r^2 + 2000 r^{-1}$

$$\frac{dA}{dr} = 2\pi r - 2000 r^{-2}$$

$$\frac{d^2A}{dr^2} = 2\pi + 4000 r^{-3}$$

A max. or min. occurs when $\frac{dA}{dr} = 0$

$$2\pi r - \frac{2000}{r^2} = 0$$

$$2\pi r^3 - 2000 = 0$$

$$\pi r^3 = 1000$$

$$r^3 = \frac{1000}{\pi}$$

$$r = \sqrt[3]{\frac{1000}{\pi}}$$

$\therefore$ Radius is 6.8 cm (2 s.f.)

when $r = \sqrt[3]{\frac{1000}{\pi}}$, $\frac{d^2A}{dr^2} > 0$

Hence a minimum (least) surface area occurs when $r = \sqrt[3]{\frac{1000}{\pi}}$